Polynomial-Time Approximability of Constrained Reinforcement Learning

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ICML International Conference On Machine Learning

Problem

For any cost criteria C, we wish to solve:

$$\max_{\pi \in \Pi} \mathbb{E}^{\pi}_{M}\left[\sum_{h=1}^{H} r_h(s_h, a_h)\right] \quad \text{s.t.} \quad \begin{cases} C^{\pi}_{M} \leq B \\ \pi \text{ deterministic} \end{cases}$$

Challenges

- Optimization is **NP-hard**
- Determining **feasibility** is NP-hard for > 1 constraint or just 1 chance constraint
- Standard approaches fail

Results

Assumption [SR]: the cost of a policy is computable recursively via a *short map* (1-Lipschitz)

Theorem: Our algorithm is a polynomial-time $(0,\epsilon)$-bicriteria for any SR criteria, which includes any constant combination of **expectation**, **almost-sure**, **anytime**, and **chance** constraints.

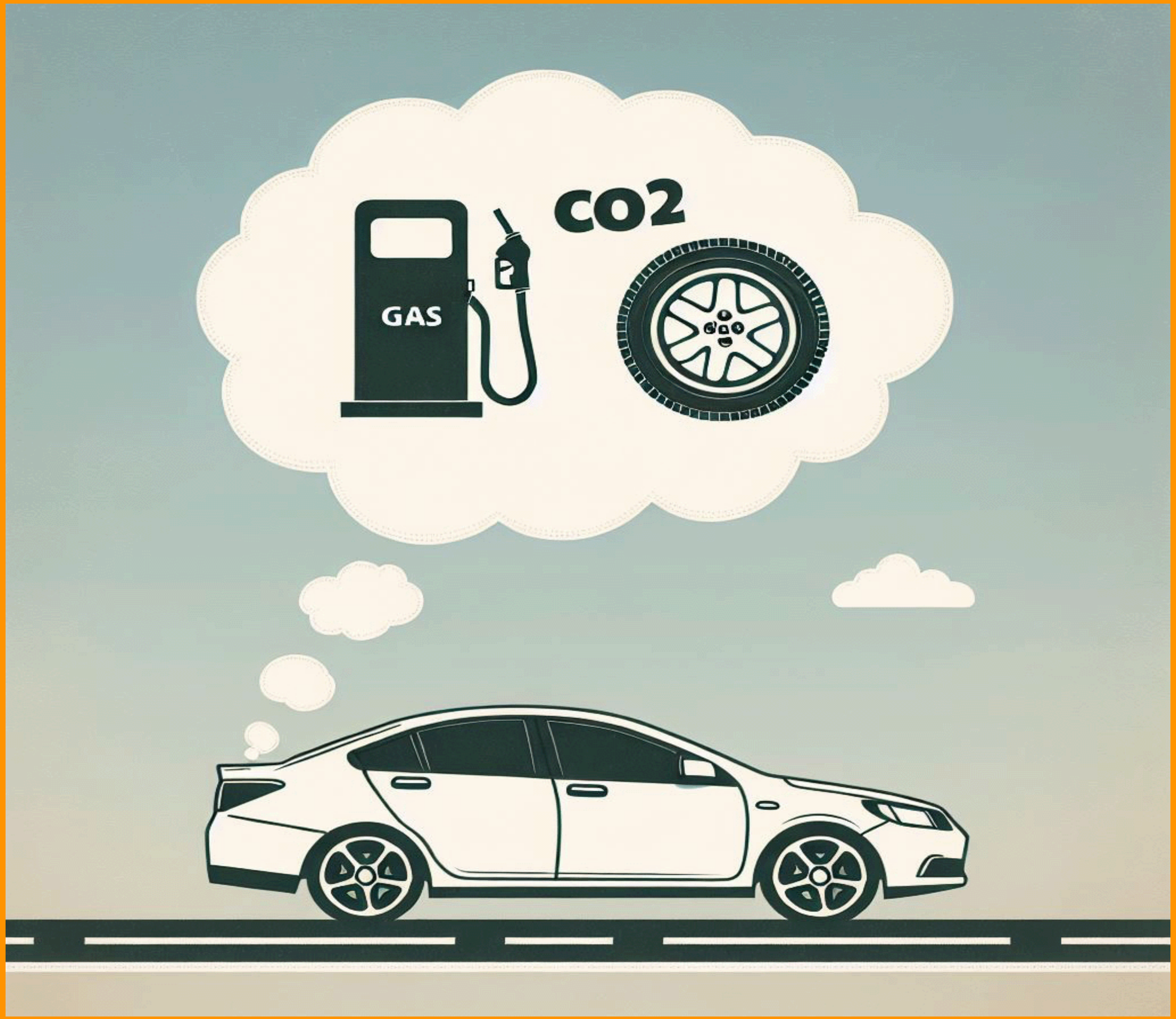
Conclusion

We establish the first ever **polynomial-time approximability** results for:

- *Chance-constrained policies*
- *Heterogenous-constrained policies*
- *Multi-expectation-constrained, deterministic policies*

Was open for 25 years!

Optimal policies satisfying nearly **any combination** of popular constraints from Constrained RL can be approximated in **polynomial-time**.



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